Ali Shafiee

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alishaffiee.github.io | Codeforces | LinkedIn | IOI Profile | Google Scholar

IOI silver medalist and ICPC World Finalist. MSc researcher in probability, stochastic control, and algorithmic game theory.

Education

Institute of Science and Technology Austria (ISTA) — Vienna, Austria
MSc in Computer Science, grade average 1.0 (Austrian scale, 1.0 highest) — *Sep. 2025 – Sep. 2027 (expected)*

Sharif University of Technology — Tehran, Iran
BSc in Computer Engineering, GPA 3.68/4 (17.03/20) — *Sep. 2018 – Aug. 2023*

Research Experience

ISTA — Vienna, Austria
Research Intern, then MSc Researcher, advisor Prof. Krishnendu Chatterjee — *Apr. 2024 – present*

- Proved NP-completeness of limit-sure reachability for small-memory POMDP policies, resolving an open complexity question; published at UAI 2025.
- Derived algorithms and complexity bounds for robust MDPs under ω-regular and discounted objectives, leading to three further papers (AAAI 2026, NeurIPS 2026, AAAI 2027).
- Implemented solvers in C++ and Python and benchmarked them against existing approaches on random and structured MDP families, scaling to instances with thousands of states.

Publications

Author lists are alphabetical, per convention in theoretical computer science.

- **Randomise Alone, Reach as a Team**. L. Brice, T. A. Henzinger, A. Montaseri, **A. Shafiee**, K. S. Thejaswini. *CAV 2026.* [paper]
- **Qualitative Analysis of ω-Regular Objectives on Robust MDPs**. A. Asadi, K. Chatterjee, E. Goharshady, M. Karrabi, **A. Shafiee**. *AAAI 2026.* [paper]
- **Limit-Sure Reachability for Small-Memory Policies in POMDPs is NP-Complete**. A. Asadi, K. Chatterjee, R. Saona, **A. Shafiee**. *UAI 2025.* [paper]
- **Quantitative Analysis of ω-Regular Robust MDPs**. A. Asadi, K. Chatterjee, E. K. Goharshady, M. Karrabi, A. Montaseri, **A. Shafiee**. *AAAI 2027, under review.* [paper]
- **On the Complexity of Discounted Robust MDPs with L_p Uncertainty Sets**. A. Asadi, K. Chatterjee, A. Montaseri, **A. Shafiee**. *NeurIPS 2026, under review.* [paper]
- **Mechanism Design for Multi-Resource Nakamoto Consensus**. N. Babashah, M. A. Baig, K. Pietrzak, **A. Shafiee**. *WINE 2026, under review.*

Competitive Programming

- **Codeforces**: **top 50 worldwide in 2019**; International Grandmaster, peak rating 2800 (handle `ToTLeS`)
- **Silver Medal, International Olympiad in Informatics (IOI)**, 2017: 76th of 300+ contestants from 80 countries
- **33rd Place, ACM-ICPC World Finals**, Moscow 2021: among 10,000+ teams from 110 countries
- **Gold Medal, ACM-ICPC West Asia Regional Contest**: 1st place 2019, 2nd 2018, 3rd 2020; 100+ teams each year
- **Silver Medal, Asia-Pacific Informatics Olympiad (APIO)**, 2018 and 2017: 20+ participating countries
- **Gold Medal, Iranian National Olympiad in Informatics**, 2016: top 8 of 10,000+ students nationally
- **Bronze Medal, Iranian Mathematical Olympiad**, 2017: top 80 of 20,000+ students nationally

Industry Experience

Quera, Iran's largest competitive-programming and technical-education platform — Tehran, Iran
Contest Manager, Content Manager, Director of CS Education, Business Line Manager — *Mar. 2019 – Aug. 2025*

- Owned a business line end to end (Oct. 2024 – Aug. 2025); previously led a 40-person division spanning six technical teams (back-end, front-end, data, blockchain, algorithms, fundamentals) and set the education product roadmap.
- Designed and shipped 12+ bootcamps and courses in C++, Python, SQL, Django, Linux/Git, data analysis, machine learning, and deep learning.
- Ran 20+ large-scale programming contests, including sponsored hiring hackathons with corporate partners.

Teaching & Service

- **Scientific Committee**, ACM-ICPC West Asia Regional Contest (2024 – present); **Host Scientific Committee**, IOI 2019, Baku
- **Problem-setter** (2017 – present) for the Iranian National Olympiad in Informatics, Codecup and Quera; **olympiad coach** (2017 – 2023), 50+ students trained for INOI and the IOI
- **Teaching Assistant**, Sharif University: Design of Algorithms, Data Structures & Algorithms, Fundamentals of Programming

Skills

Programming: C++, Python, Rust · **Numerical & ML**: NumPy, SciPy, pandas, Matplotlib, scikit-learn, PyTorch, JAX
Tools: Git, Linux/Bash, SQL, Docker, Jupyter, LaTeX · **Mathematics**: probability, stochastic processes, Markov decision processes, optimization, game theory
Languages: Persian (native), English (TOEFL 110/120)